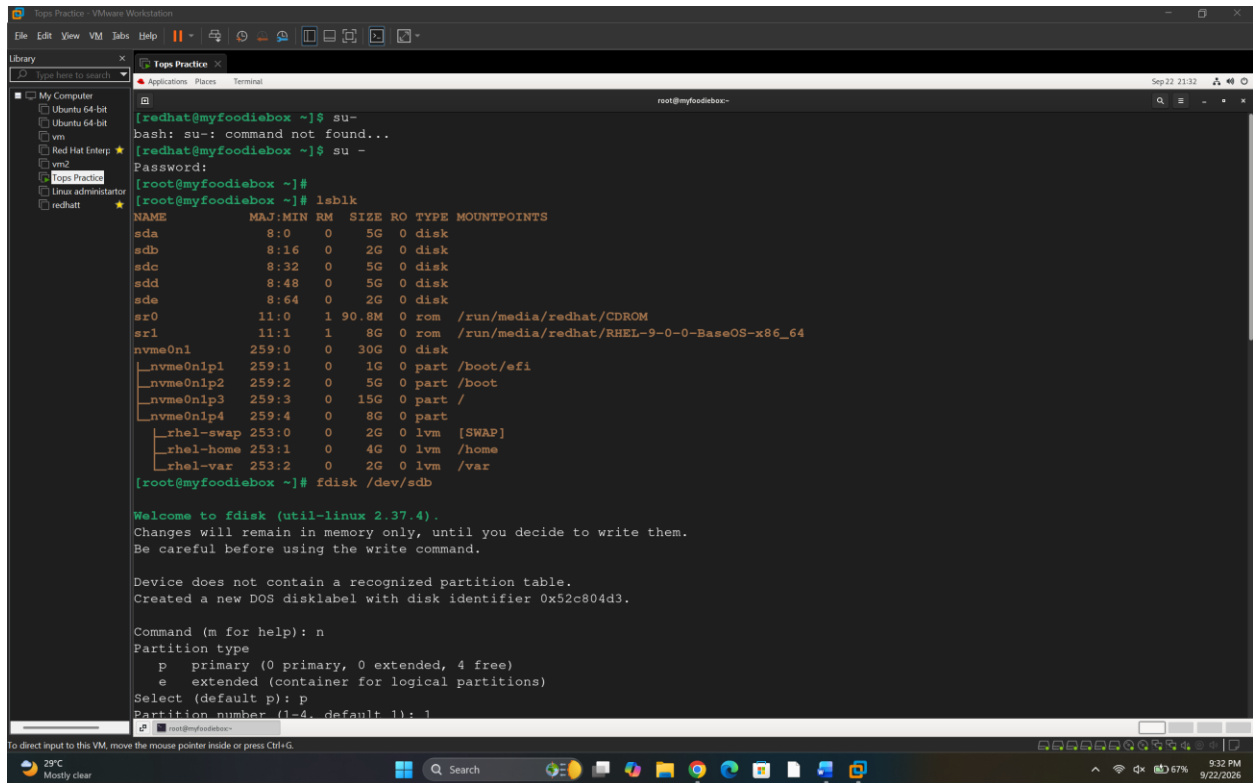


Tasks

1.

Create two 200MB partitions using fdisk or parted, then use the lsblk command to verify their creation.

>



```
[redhat@myfoodiebox ~]$ su-
bash: su-: command not found...
[redhat@myfoodiebox ~]$ su -
Password:
[root@myfoodiebox ~]# lsblk
[redhat@myfoodiebox ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x52c804d3.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	5G	0	disk	
sdb	8:16	0	2G	0	disk	
sdc	8:32	0	5G	0	disk	
sdd	8:48	0	5G	0	disk	
sde	8:64	0	2G	0	disk	
sr0	11:0	1	90.8M	0	rom	/run/media/redhat/CDROM
sr1	11:1	1	8G	0	rom	/run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1	259:0	0	30G	0	disk	
nvme0n1p1	259:1	0	1G	0	part	/boot/efi
nvme0n1p2	259:2	0	5G	0	part	/boot
nvme0n1p3	259:3	0	15G	0	part	/
nvme0n1p4	259:4	0	8G	0	part	
rhel-swap	253:0	0	2G	0	lvm	[SWAP]
rhel-home	253:1	0	4G	0	lvm	/home
rhel-var	253:2	0	2G	0	lvm	/var

```
Tops Practice - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp
  vm2
  Tops Practice
  Linux administrator
  redhat
Applications Places Terminal
root@myfoodiebox:~# fdisk /dev/sdb

Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048): 2048
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303): 200M+
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303): +200M

Created a new partition 1 of type 'Linux' and of size 200 MiB.

Command (m for help): W
W: unknown command

Command (m for help): w

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@myfoodiebox ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): n
Partition type
  p   primary (1 primary, 0 extended, 3 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (411648-4194303, default 411648): 411648
Last sector, +/-sectors or +/-size(K,M,G,T,P) (411648-4194303, default 4194303): +200M

Created a new partition 2 of type 'Linux' and of size 200 MiB.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G
29°C
Mostly clear
9:32 PM
9/22/2026
```

```
Tops Practice - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp
  vm2
  Tops Practice
  Linux administrator
  redhat
Applications Places Terminal
root@myfoodiebox:~# fdisk /dev/sdb

Partition type
  p   primary (1 primary, 0 extended, 3 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (411648-4194303, default 411648): 411648
Last sector, +/-sectors or +/-size(K,M,G,T,P) (411648-4194303, default 4194303): +200M

Created a new partition 2 of type 'Linux' and of size 200 MiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@myfoodiebox ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   5G  0 disk
sdb          8:16   0   2G  0 disk
└─sdb1       8:17   0  200M 0 part
└─sdb2       8:18   0  200M 0 part
sdc          8:32   0   5G  0 disk
sdd          8:48   0   5G  0 disk
sde          8:64   0   2G  0 disk
sr0         11:0    1 90.8M  0 rom  /run/media/redhat/CDROM
srl         11:1    1    8G  0 rom  /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1     259:0    0   30G  0 disk
└─nvme0n1p1 259:1    0    1G  0 part /boot/efi
└─nvme0n1p2 259:2    0    5G  0 part /boot
└─nvme0n1p3 259:3    0   15G  0 part /
└─nvme0n1p4 259:4    0    8G  0 part
└─rhel-swap 253:0    0    2G  0 lvm  [SWAP]
└─rhel-home 253:1    0    4G  0 lvm  /home
└─rhel-var  253:2    0    2G  0 lvm  /var

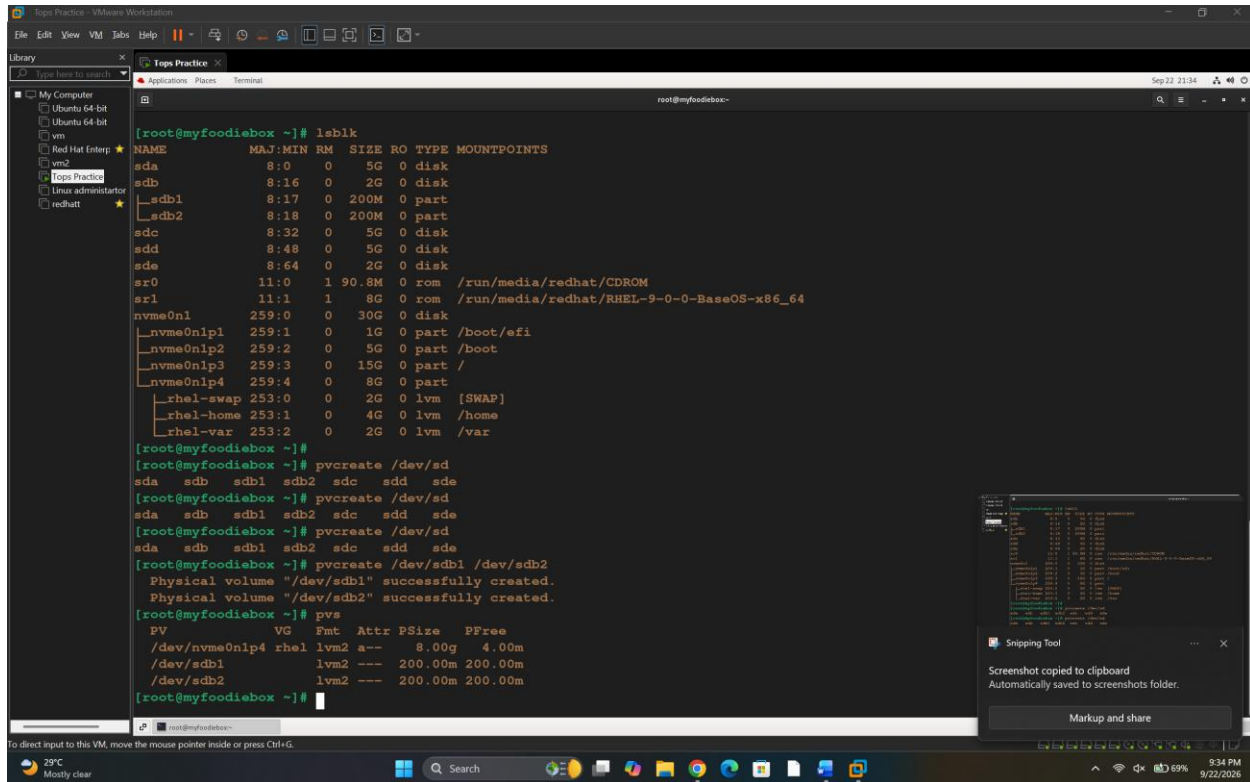
[root@myfoodiebox ~]#
[root@myfoodiebox ~]#

To direct input to this VM, move the mouse pointer inside or press Ctrl+G
29°C
Mostly clear
9:32 PM
9/22/2026
```

2.

Initialize both partitions as physical volumes using the pvcreate command and confirm their status with pvs.

>



The screenshot shows a terminal window in a VM named 'myfoodiebox'. The user has run the 'lsblk' command, which displays a list of disk partitions. The partitions are as follows:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	5G	0	disk	
sdb	8:16	0	2G	0	disk	
sdb1	8:17	0	200M	0	part	
sdb2	8:18	0	200M	0	part	
sdc	8:32	0	5G	0	disk	
sdd	8:48	0	5G	0	disk	
sde	8:64	0	2G	0	disk	
sr0	11:0	1	90.8M	0	rom	/run/media/redhat/CDROM
sr1	11:1	1	8G	0	rom	/run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1	259:0	0	30G	0	disk	
nvme0n1p1	259:1	0	1G	0	part	/boot/efi
nvme0n1p2	259:2	0	5G	0	part	/boot
nvme0n1p3	259:3	0	15G	0	part	/
nvme0n1p4	259:4	0	8G	0	part	
rhel-swap	253:0	0	2G	0	lvm	[SWAP]
rhel-home	253:1	0	4G	0	lvm	/home
rhel-var	253:2	0	2G	0	lvm	/var

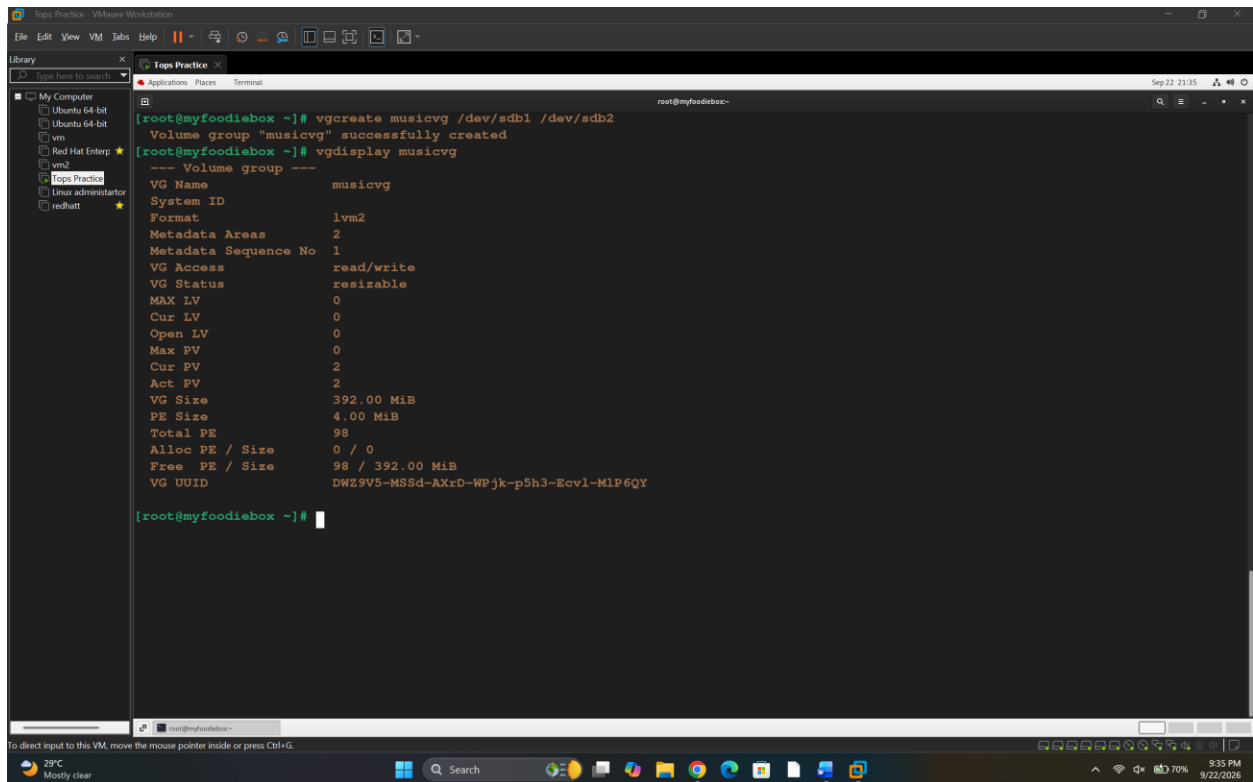
The user then runs the 'pvcreate' command on the partitions /dev/sdb1 and /dev/sdb2. The output shows that the physical volumes were successfully created. Finally, the user runs the 'pvs' command, which displays the following information:

PV	VG	Fmt	Attr	PSize	PFree
/dev/nvme0n1p4	rhel	lvm2	a--	8.00g	4.00m
/dev/sdb1		lvm2	---	200.00m	200.00m
/dev/sdb2		lvm2	---	200.00m	200.00m

3.

Create a new volume group named musicvg using the two physical volumes, then display its details with vgdisplay.

>



4.

Create a logical volume called playlistlv of size 300MB inside musicvg, format it with ext4, and mount it to /mnt/playlist.

>

```

[root@myfoodiebox ~]# lvcreate -L 300M -n playlistlv musicvg
Logical volume "playlistlv" created.
[root@myfoodiebox ~]# lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
playlistlv musicvg -wi-a---- 300.00m
home rhel -wi-ao---- 4.00g
swap rhel -wi-ao---- 2.00g
var rhel -wi-ao---- 2.00g
[root@myfoodiebox ~]# mkfs.ext4 /dev/musicvg/playlistlv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: c9e62a73-e54d-47f2-a943-f0e0af58c7f2
Superblock backups stored on blocks:
8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@myfoodiebox ~]# mkdir -p /mnt/playlist
[root@myfoodiebox ~]# mount /dev/musicvg/playlistlv /mnt/playlist
[root@myfoodiebox ~]# df -h /mnt/playlist
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/musicvg-playlistlv 272M 14K 253M 1% /mnt/playlist
[root@myfoodiebox ~]# lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 5G 0 disk
sdb 8:16 0 2G 0 disk
sdb1 8:17 0 200M 0 part
└─musicvg-playlistlv 253:3 0 300M 0 lvm /mnt/playlist
sdb2 8:18 0 200M 0 part
└─musicvg-playlistlv 253:3 0 300M 0 lvm /mnt/playlist
sdc 8:32 0 5G 0 disk
sdd 8:48 0 5G 0 disk
sde 8:64 0 2G 0 disk
```

```

8193, 24577, 40961, 57345, 73729, 204801, 221185
Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@myfoodiebox ~]# mkdir -p /mnt/playlist
[root@myfoodiebox ~]# mount /dev/musicvg/playlistlv /mnt/playlist
[root@myfoodiebox ~]# df -h /mnt/playlist
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/musicvg-playlistlv 272M 14K 253M 1% /mnt/playlist
[root@myfoodiebox ~]# lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 5G 0 disk
sdb 8:16 0 2G 0 disk
sdb1 8:17 0 200M 0 part
└─musicvg-playlistlv 253:3 0 300M 0 lvm /mnt/playlist
sdb2 8:18 0 200M 0 part
└─musicvg-playlistlv 253:3 0 300M 0 lvm /mnt/playlist
sdc 8:32 0 5G 0 disk
sdd 8:48 0 5G 0 disk
sde 8:64 0 2G 0 disk
sr0 11:0 1 90.8M 0 rom /run/media/redhat/CDROM
sr1 11:1 1 8G 0 rom /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1 259:0 0 30G 0 disk
└─nvme0n1p1 259:1 0 1G 0 part /boot/efi
└─nvme0n1p2 259:2 0 5G 0 part /boot
└─nvme0n1p3 259:3 0 15G 0 part /
└─nvme0n1p4 259:4 0 8G 0 part
└─rhel-swap 253:0 0 2G 0 lvm [SWAP]
└─rhel-home 253:1 0 4G 0 lvm /home
└─rhel-var 253:2 0 2G 0 lvm /var

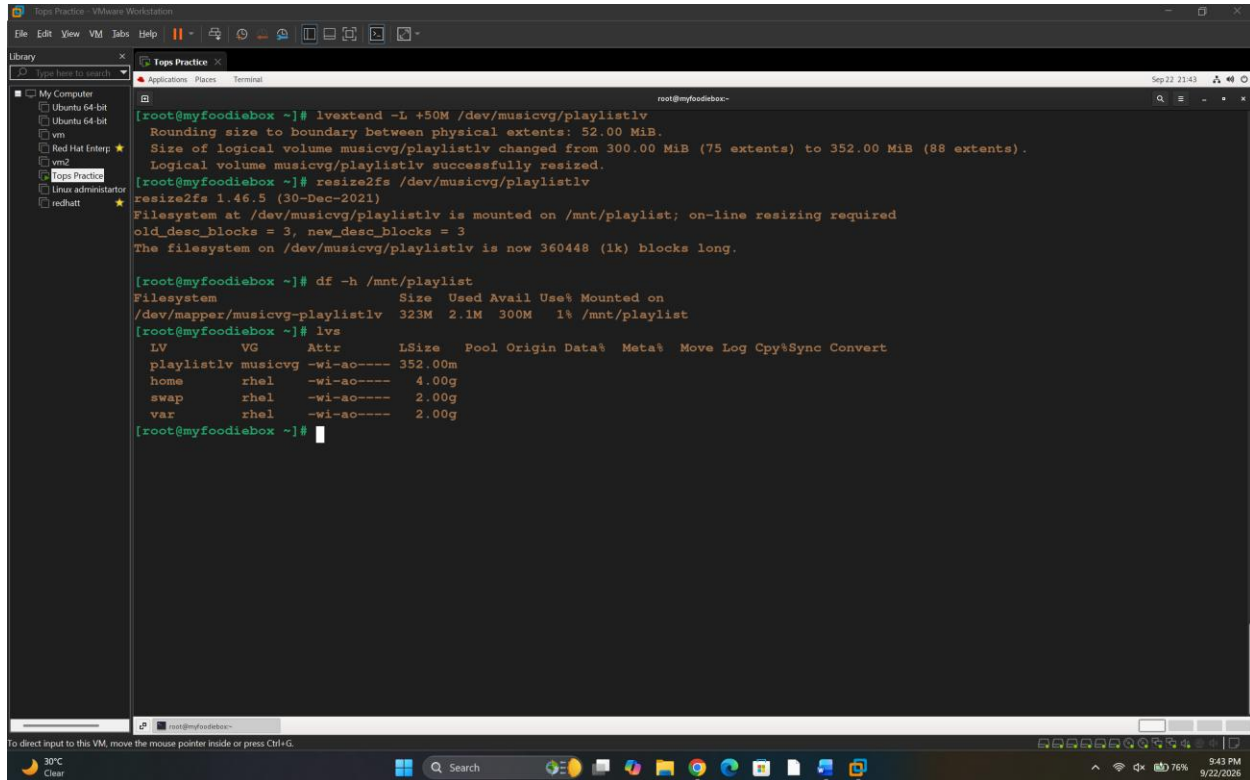
[root@myfoodiebox ~]#
[root@myfoodiebox ~]#
[root@myfoodiebox ~]#
```

5.

Resize playlistlv by adding 50MB more space using lvextend, then expand the filesystem and show the updated size with df -h.

Hint: Use resize2fs after lvextend to update the filesystem size.

>



```
[root@myfoodiebox ~]# lvextend -L +50M /dev/musicvg/playlistlv
Rounding size to boundary between physical extents: 52.00 MiB.
Size of logical volume musicvg/playlistlv changed from 300.00 MiB (75 extents) to 352.00 MiB (88 extents).
Logical volume musicvg/playlistlv successfully resized.
[root@myfoodiebox ~]# resize2fs /dev/musicvg/playlistlv
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/musicvg/playlistlv is mounted on /mnt/playlist; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 3
The filesystem on /dev/musicvg/playlistlv is now 360448 (1k) blocks long.

[root@myfoodiebox ~]# df -h /mnt/playlist
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/musicvg-playlistlv 323M    2.1M   300M   1% /mnt/playlist
[root@myfoodiebox ~]# lvs
LV          VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
playlistlv  musicvg -wi-ao---- 352.00m
home        rhel    -wi-ao---- 4.00g
swap        rhel    -wi-ao---- 2.00g
var         rhel    -wi-ao---- 2.00g
[root@myfoodiebox ~]#
```